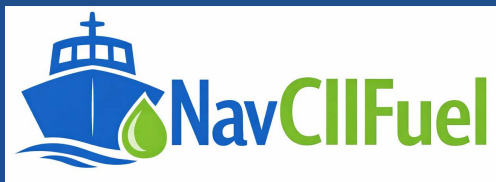


MARITIME COMPLIANCE TOOLS

USER GUIDE

4-Stroke Engine —SFOC Correction Calculator — Web Version



Specific Fuel Oil Consumption Correction · Web Version

Prepared for: Vessel Compliance & Reporting Teams

Document version: 1.0

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1. Introduction

1.1 Purpose of this guide

This guide explains how to use the 4-Stroke Engine — ISO 3046 SFOC Correction Calculator, a browser-based tool that corrects Specific Fuel Oil Consumption (SFOC) measurements from actual site (test) conditions to ISO 3046-1 standard reference conditions. It walks through every input field, the engine data table, the calculated results, the fleet-average summary and comparison charts.

The guide is written for marine engineers, sea-trial teams and performance-testing personnel who need to enter measured fuel-consumption data, run the ISO correction, and read or export the corrected SFOC results.

1.2 What the tool does

- Implements the ISO 3046-1 standard for correcting SFOC measurements from actual site conditions to standard reference conditions.
- Supports up to 3 engines simultaneously, each with its own independent data table.
- Computes the ISO correction factors k (ambient), α (alfa) and β (beta), and applies the fuel calorific-value correction.
- Calculates the final ISO 3046 corrected SFOC (g/kWh) for each engine load point (25/50/75/100%).
- Produces a fleet-averaged summary across all active engines, plus per-engine and combined comparison charts (SFOC Measured vs SFOC ISO Corrected).

Before you start: Have the engine test/sea-trial data ready — fuel mass measured (A), measurement duration (C), ambient air temperature (Tx), and charge-air coolant temperature (Tcx) for each load point — together with the engine's MCR and, if available, its maker's test-certificate constants (buoyancy factor, NCV values, mechanical efficiency, correction exponents). The tool runs entirely in your browser; use a current version of Chrome, Edge or Firefox.

2. Interface Overview

The calculator is organised into a fixed top toolbar, an ISO 3046 reference strip, a load-points control, one tab per engine (up to 3), and — after calculating — a fleet-average summary with comparison charts.



Figure 1 — Toolbar: Back / Home navigation, tool title, and the User Guide button.

The toolbar is available at all times. Back and Home provide page navigation; the User Guide button (top right) opens the tool's built-in, on-screen help panel — which contains the same step-by-step instructions, parameter table and formulas covered in this document.

3. Quick Start Workflow

Follow these steps for a typical single- or multi-engine SFOC correction:

Step	What to do
1	Open the calculator HTML file in your browser. Confirm the ISO 3046 Reference strip and Load Points controls are visible at the top of the page.
2	Review the ISO 3046 Reference values (pr, Tr, fir, Tcr). Leave at the ISO defaults unless your certificate specifies otherwise.
3	Use + Add Load / × on a load chip to match the load points you tested (25%, 50%, 75%, 100%).
4	Click + Add Engine to create up to 3 engine tabs. Enter Engine No., MCR [kW] and an optional Order No. for each.
5	In the white input rows, enter the measured values (A, C, Tx, Tcx) for each active load column, from the test/sea-trial record.
6	Review the green default rows (B, G, H, ηm, m, n, s, l) and override with values from the engine maker's test report where available.
7	Click Calculate on each engine panel. Required fields left empty are highlighted in red.
8	Read the blue intermediate rows and the yellow SFOC ISO Corrected result row. Review the Fleet Average summary and comparison charts.

4. ISO 3046 Reference & Load Points

Section at the top of the page: the ISO 3046 Reference strip defines the standard atmosphere used for every correction, and the Load Points control defines which engine-load columns appear in the data table.

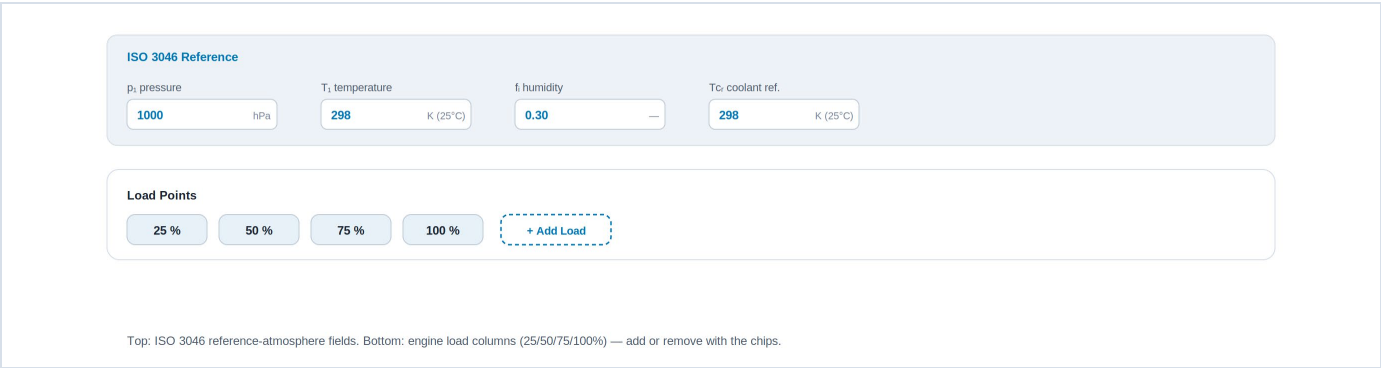


Figure 2 — ISO 3046 Reference strip (top) and Load Points control (bottom).

4.1 ISO 3046 Reference fields

Field	Meaning	ISO default
pr	Reference barometric pressure	1000 hPa
Tr	Reference air temperature	298 K (25°C)
fir	Reference relative humidity	0.30
Tcr	Reference charge-air coolant temperature	298 K (25°C)

Change these only if the engine's test certificate specifies non-standard reference conditions; otherwise leave at the ISO 3046-1 defaults shown above.

4.2 Load Points

Use + Add Load to add a load column (from the standard set 25/50/75/100%), or click × on a load chip to remove one. The engine data table, the fleet-average summary and the comparison charts update automatically to match the active load columns.

5. Engines & the Data Table

Click + Add Engine to create up to 3 independent engine tabs, each with its own data table, MCR and Order No. Switch tabs to move between engines; click × on the active tab to remove that engine.

The screenshot shows the 'Engine 1' tab selected. At the top, there are buttons for 'Engine 1', 'Engine 2', and '+ Add Engine'. Below these are input fields for 'Engine No.' (1), 'MCR [kW]' (2600), and 'Order No. (optional)' (optional). The main data table has columns for 'Parameter', '25%', '50%', '75%', '100%', 'Unit', and 'Description'. The rows are color-coded: white for measurement inputs (A, B, k), green for editable default constants, blue for intermediate calculation steps (SFOC ISO Corrected), and yellow for the final ISO 3046 corrected result. Below the table are 'Calculate' and 'Reset defaults' buttons. A 'Row Color Legend' explains the color coding: White for measurement inputs, Green for editable default constants, Blue for intermediate calculation steps, and Yellow for the final ISO 3046 corrected result. A note at the bottom states: 'Each engine tab has its own table. Rows are grouped: Measurement Inputs, Ambient Conditions, Correction Exponents, SFOC Measurements, ISO Correction Factors and Results.'

Parameter	25%	50%	75%	100%	Unit	Description
MEASUREMENT INPUTS						
A					kg	Fuel measured
B					—	Buoyancy correction factor
k	—	—	—	—	—	Ambient correction factor (auto)
SFOC ISO Corrected	183.4	176.1	172.9	174.5	g/kWh	Final ISO 3046 corrected result

Row Color Legend

- White — measurement inputs (enter from test)
- Green — editable default constants
- Blue — intermediate calculation steps (auto)
- Yellow — final ISO 3046 corrected result

Each engine tab has its own table. Rows are grouped: Measurement Inputs, Ambient Conditions, Correction Exponents, SFOC Measurements, ISO Correction Factors and Results.

Figure 3 — Engine panel: tabs, meta fields, data table and the row color legend.

5.1 Engine meta fields

Field	Description
Engine No.	Sequential identifier for the engine (1, 2, 3...).
MCR [kW]	Maximum Continuous Rating — required. Used to compute generator output D and each load point's SFOC.
Order No.	Optional free-text reference for the engine/contract; does not affect calculations.

5.2 Row color coding

Color	Meaning
White	Direct measurement inputs — must be entered from the test (A, C, Tx, Tcx).
Green	Editable default constants, pre-filled from typical ISO 3046-1 / engine-maker values (B, G, H, η_m , m, n, s, l).
Blue	Intermediate ISO correction steps, computed automatically (D, SFOC Measured, k, α , β).

Color	Meaning
Yellow	Final SFOC ISO Corrected result (g/kWh) — read-only output.

Click Calculate on an engine's panel once its required (white and green) fields are complete — a field left empty is highlighted in red. Use Reset defaults to restore only that engine's green-row constants; this does not affect other engines or any entered white-row measurements.

6. Input Parameter Reference

Full list of parameters used across the data table, in the order they appear:

Symbol	Unit	Type	Description
A	kg	Input	Fuel mass measured during the test period
B	—	Default	Buoyancy correction factor (typically 0.9944 for marine diesel)
C	s	Input	Measurement duration in seconds
D	kW	Calculated	Generator output — auto: $MCR \times \text{load\%} \times \eta_g$
G	MJ/kg	Default	Net calorific value (NCV) of the actual fuel used
H	MJ/kg	Default	Standard NCV reference value (ISO standard fuel)
T _x	°C	Input	Ambient air temperature at turbocharger inlet
P _x	hPa	Default	Ambient total barometric pressure at site
T _{cx}	°C	Input	Charge air coolant temperature at engine inlet
η_m	—	Default	Mechanical efficiency (use 0.8 if unknown; from maker's data)
m	—	Default	Pressure correction exponent (ISO Table 2)
n	—	Default	Temperature correction exponent
s	—	Default	Charge air coolant correction exponent
I (η_g)	—	Default	Generator efficiency (typically 0.96 for 4-stroke AEs)
SFOC _m	g/kWh	Calculated	Measured SFOC — auto from A, B, C, MCR, load%
k	—	Calculated	Ambient correction factor (pressure $\times$ temperature $\times$ coolant)
α (alfa)	—	Calculated	Corrected factor with mechanical efficiency applied
β (beta)	—	Calculated	k / α ratio used in the final correction
SFOCLCV	g/kWh	Calculated	SFOC corrected for fuel calorific value: $SFOC_m \times G/H$
SFOCISO	g/kWh	Result	Final ISO 3046 corrected SFOC: $(SFOCLCV / \beta) \times \eta_g$

7. Calculation Formulas (ISO 3046-1)

The calculator applies the following ISO 3046-1 formulas automatically; no manual calculation is required.

SFOC Measured $SFOC_m = (B \times A \times 1000 / C) \times 3600 / (\text{load\%} \times MCR)$

Ambient correction factor k $k = (P_x / p_r)^m \times (T_r / (T_x + 273.15))^n \times (T_{cr} / (T_{cx} + 273.15))^s$

Alpha (α) — mechanical efficiency correction $\alpha = k - 0.7 \times (1 - k) \times (1/\eta_m - 1)$

Beta (β) $\beta = k / \alpha$

SFOC LCV corrected $SFOCLCV = SFOC_m \times G / H$

SFOC ISO Corrected — final result $SFOCISO = (SFOCLCV / \beta) \times \eta_g$

P_x , T_x and T_{cx} are the site (measured) conditions; p_r , T_r and T_{cr} are the ISO reference values entered in the top strip. T_x and T_{cx} are entered in °C and converted internally to Kelvin (+273.15); T_r and T_{cr} in the top strip are already in Kelvin.

8. Results — Fleet Average Summary & Charts

Once Calculate has been run for the active engines, the Fleet Average summary and the SFOC comparison charts appear below the engine panels.

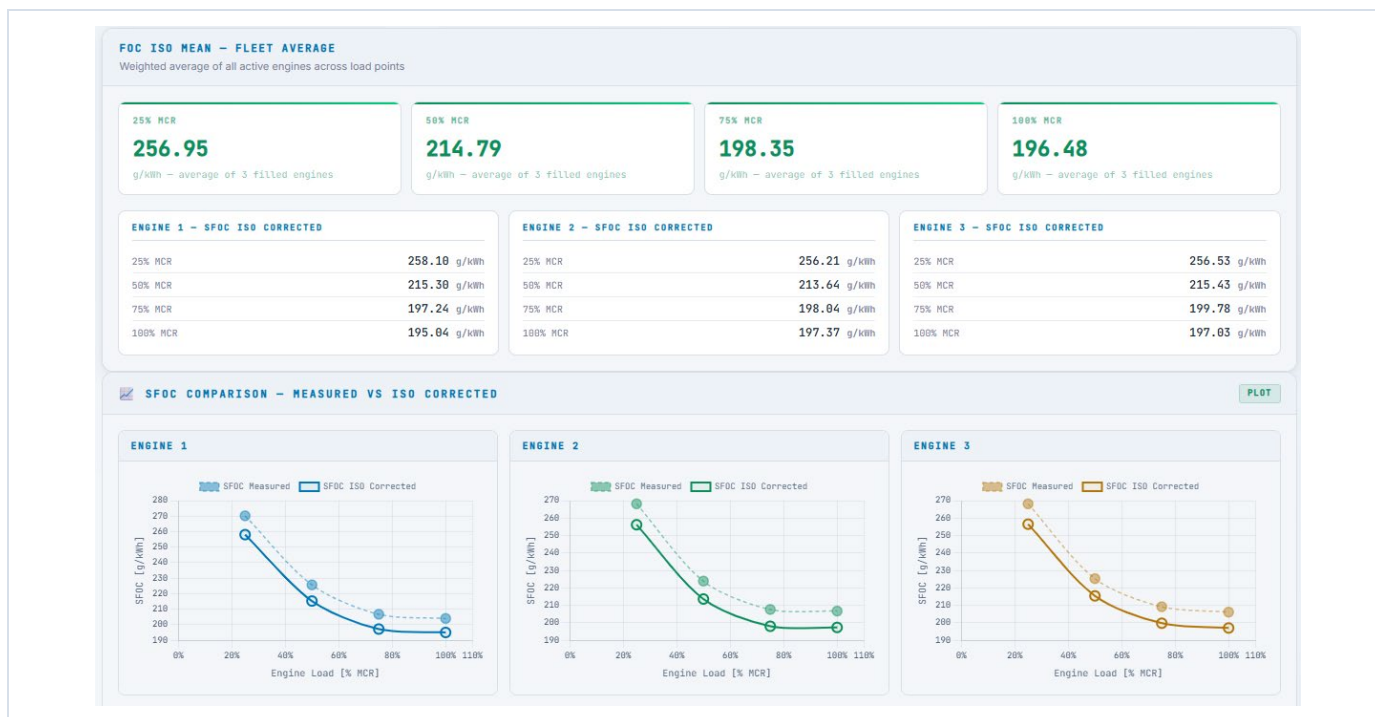


Figure 4 — Fleet Average summary cards and the SFOC Measured vs ISO Corrected comparison chart.

Section	Meaning
FOC ISO Mean — Fleet Average	Arithmetic mean of the ISO-corrected SFOC across all active engines, shown for each load point. This is the standard method for fleet performance reporting.
Per-engine mini charts	One small chart per engine, plotting SFOC Measured against SFOC ISO Corrected across the active load points.
Combined — ISO Corrected SFOC, all engines	Overlay chart comparing the ISO-corrected SFOC curve of every active engine on one plot, for direct comparison.

9. Best-Practice Checklist

- Ensure all measured values (A, C, Tx, Tcx, Px) come from calibrated instruments at stable, steady-state load conditions during the test.
- Enter Tx and Tcx in °C — the calculator converts to Kelvin internally; do not enter Kelvin directly in these fields.
- Confirm MCR is entered correctly for each engine before calculating — it is used to derive generator output D and every SFOC value.
- Review the green default constants against the engine maker's test certificate rather than relying on the pre-filled ISO defaults where certified values are available.
- Use Reset defaults only when you want to restore one engine's green-row constants; it will not clear your measured (white-row) inputs.
- The downloaded HTML file is self-contained (Chart.js included) and keeps all entered data — it can be opened offline, emailed, or printed as a record.

10. Troubleshooting

Symptom	Likely cause / fix
Field is highlighted red after clicking Calculate	A required white or green field is empty for that engine. Fill in the highlighted field and click Calculate again.
Fleet Average / charts do not appear	Calculate has not been run successfully for at least one engine, or all required fields are still empty.
Load column values look wrong after adding a load	New load columns start blank (or copy the nearest existing column) — re-enter the measured values for that column before calculating.
Removed the wrong engine	Engines are not saved between sessions unless the file itself is saved; re-enter the engine's data or reload a previously saved copy of the file.
Chart legend or combined chart missing an engine	Only engines with a completed Calculate are included in the fleet summary and combined chart — calculate that engine's panel.

Appendix A — Full Interface Reference

The figures below summarise the complete calculator layout, for quick reference alongside the section-by-section descriptions in this guide.

4 STROKE ENGINE - SFOC CORRECTION CALCULATOR

ISO 3046 REFERENCE P_k PRESSURE 1000 n_{Pu} T_g TEMPERATURE 298 K (20°C) F_{th} HUMIDITY 0.3 T_{cg} COOLANT REF. 298 K (25°C)

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Toolbar

LOAD POINTS Add or remove engine load columns — 25%, 50%, 75%, 100%

25 % x 50 % x 75 % x 100 % x + Add Load Use + Add Load or x to update columns, summary and charts.

ENGINES Select engine to enter data — up to 3 engines, FOC mean averaged across all

Engine 1 Engine 2 x Engine 3

ENGINE NO. MCR [KW] ORDER NO.

2 510 optional

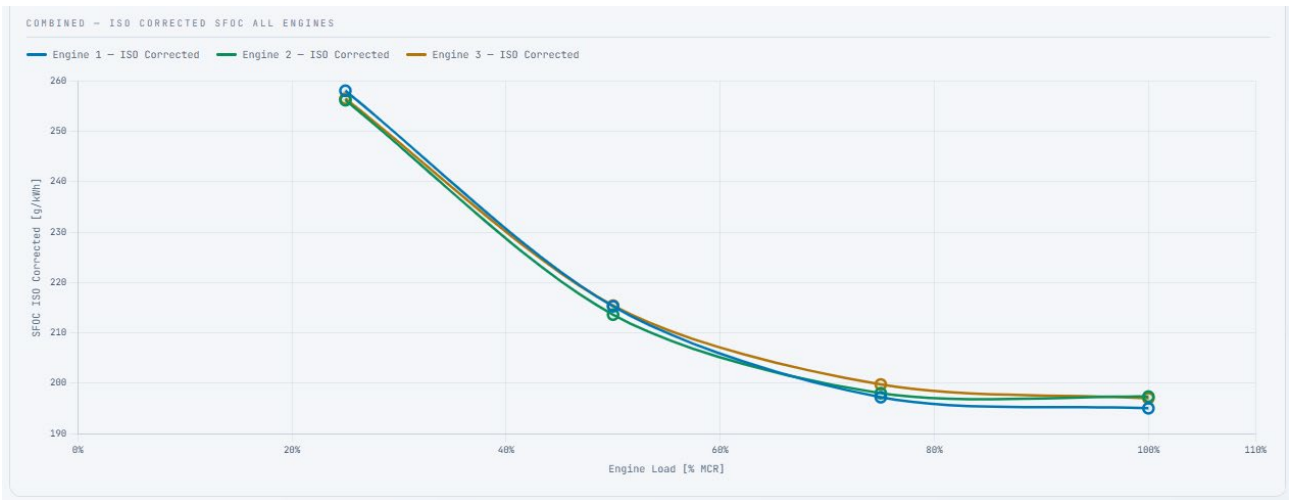
☐ Input fields ☒ Default values (can be changed) ☐ Calculation steps ☐ Result

ISO 3046 Reference & Load Points

PARAMETER	ENGINE LOAD				UNIT	DESCRIPTION
	25%	50%	75%	100%		
MEASUREMENT INPUTS						
A	1	1	1.5	2	kg	Fuel measured
B	0.9944	0.9944	0.9944	0.9944	—	Buoyancy correction factor
C	104.7	62.7	67.6	67.9	s	Measured time — Measurement duration in seconds
D	122.40	244.80	367.20	489.60	kW	Generator output — auto-calculated (MCR × load × η_g)
G	42.62	42.62	42.62	42.62	kJ/kg	Net calorific value (NCV) of fuel
H	42.7	42.7	42.7	42.7	kJ/kg	Standard net calorific value
AMBIENT CONDITIONS ON SITE						
Tx	25	29	30	26	°C	Ambient air temperature at turbocharger inlet
Px	1010.3	1010.3	1010.3	1010.3	hPa	Ambient total barometric pressure
Tcx	38	38	39	40	°C	Charge air coolant temperature
CORRECTION EXPONENTS & EFFICIENCIES						
η_m	0.9	0.9	0.9	0.9	—	Mechanical efficiency (use 0.8 if unknown)
m	0.7	0.7	0.7	0.7	—	Exponent — pressure adjustment (ISO Table 2)
n	1.2	1.2	1.2	1.2	—	Exponent — temperature adjustment
s	1	1	1	1	—	Exponent — charge air coolant
SFOC MEASUREMENTS						
SFOC Measured	268.1679	223.9009	207.6714	206.7539	g/kWh	Auto-calculated from A, B, C, MCR — formula: $B \times A^{1000/C} \times 3600 / (\text{load} \times \text{MCR})$
ISO 3046 CORRECTION FACTORS (CALCULATED)						
k	0.964050	0.948755	0.941973	0.954051	—	Ambient correction factor: $(P_x/P)^m \times (T/T_x)^n \times (T_x/T_{cx})^s$
α (alfa)	0.961254	0.944769	0.937460	0.950478	—	Corrected factor with mechanical efficiency: $k - 0.7 \times (1-k) \times (\eta_m - 1)$
β (beta)	1.002909	1.004219	1.004814	1.003760	—	k / α
SFOC RESULTS						
$I(\eta_g)$	0.96	0.96	0.96	0.96	—	Generator efficiency
SFOC LCV	267.6655	223.4814	207.2823	206.3665	g/kWh	SFOC corrected for fuel calorific value: $\text{SFOC}_{LCV} \times G/H$
SFOC ISO Corrected	256.21	213.64	198.04	197.37	g/kWh	ISO 3046 corrected result: $(\text{SFOC}_{LCV} / \beta) \times \eta_g$

Calculate Reset defaults

Engine tabs, data table & row color legend



Fleet Average summary & comparison chart

End of user guide.